
Breast Cancer Detection System - WSI Analysis

Automated Detection Using Fine-tuned CellViT with Hybrid Pooling

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MODEL ARCHITECTURE

Base Model: CellViT (Cell Vision Transformer)

- A hybrid CNN-Transformer architecture for computational pathology
- Pre-trained on PanNuke dataset (200,000+ labeled nuclei)
- Fine-tuned on breast cancer histopathology slides

Configuration:

- _ num_nuclei_classes: 6 (nucleus type classification)
- _ num_tissue_classes: 19 (tissue region classification)
- _ embed_dim: 384 (embedding dimension)
- _ depth: 12 (transformer layers)
- _ num_heads: 6 (multi-head attention)
- _ extract_layers: [3,6,9,12] (feature extraction layers)

FINE-TUNING DETAILS

Model Checkpoint: cellvit_breast_final.pth
Training Data: 200 breast histopathology slides (137 train / 63 val)
Task: Binary classification (Benign vs Malignant)

INFERENCE PIPELINE

1. Load fine-tuned CellViT model
2. Load external test patches and slide-level ground truth
3. Group patches by source slide (~200 patches/slide)
4. Generate patch-level malignancy probabilities
5. Apply Hybrid Pooling (Top-K + Mean, $\alpha=0.7$, $K=100$)
6. Compare with baseline pooling methods (Mean, Top-K)
7. Calculate comprehensive metrics:
 - Accuracy, Precision, Recall, F1-Score
 - ROC Curve and AUC
 - Confusion Matrix
8. Analyze error cases (False Negatives, False Positives)
9. Export results and visualizations

HYBRID POOLING STRATEGY

Formula: $\text{Slide_Score} = \alpha \times \text{Top-K_Mean} + (1-\alpha) \times \text{All_Mean}$

Parameters:

- K: 100 (top suspicious patches)
- α : 0.7 (weight given to Top-K)

- Threshold: 0.5 (binary decision boundary)

Rationale:

- Top-K focuses on highly suspicious regions
- Mean provides global context
- Hybrid balances sensitivity and specificity

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OUTPUTS

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- roc_curve_hybrid_pooling.png : ROC curve visualization
 - results_hybrid_pooling.pkl : Complete results dictionary
 - Console output : Detailed metrics and error analysis
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Device: cuda

Loading pre-trained CellViT model...

✓ Model loaded

Loading external test data (86 slides)...

Loaded 17200 patches from 86 slides

Malignant slides: 11000

Benign slides: -10914

Predicting with Hybrid Pooling (K=100, $\alpha=0.7$)...

Processing slides: 100%|██████████| 86/86 [14:47<00:00, 10.32s/it]

Applying threshold = 0.5...

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RESULTS - HYBRID POOLING (Top-K + Mean)

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POOLING SETTINGS:

- └ K (Top patches): 100
- └ Alpha (Top-K weight): 0.7
- └ Threshold: 0.5

SLIDE-LEVEL METRICS:

- └ Accuracy: 88.37%
- └ Precision: 90.91%
- └ Recall: 90.91%
- └ F1-Score: 0.909
- └ AUC: 0.9179

CONFUSION MATRIX:

- └ TN (Benign → Benign): 26
- └ FP (Benign → Malignant): 5 False Alarm
- └ FN (Malignant → Benign): 5 Missed Cancer
- └ TP (Malignant → Malignant): 50

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FALSE NEGATIVES (Missed Cancers)

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5 malignant slides were missed:

#	Slide Name	Hybrid	Top-10	All Mean
1	TCGA-E9-A2JS-01A-01-TSA.8C89FED2-C22D-4C19-990B-22FD44EC8AEB.svs	47.69%	48.41%	46.00%
2	TCGA-C8-A133-01A-03-BSC.cbc07c65-6e59-401f-aa06-549c488cf575.svs	47.64%	50.10%	41.91%
3	TCGA-D8-A1X7-01A-01-TS1.79381377-01d0-4296-abec-b522c078559c.svs	41.32%	43.68%	35.82%
4	TCGA-C8-A1HM-01A-01-TSA.8a4cc75b-80ec-45e3-8b1a-a5e66aea10b5.svs	32.80%	36.89%	23.26%
5	TCGA-AC-A5XU-01A-01-TSA.53BFAB63-FB19-4F47-B71C-C93645D58BBB.svs	31.32%	32.91%	27.60%

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FALSE POSITIVES (False Alarms)

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5 bening slides flagged as malignant:

#	Slide Name	Hybrid	Top-10	All Mean
1	TCGA-BH-A0DQ-11A-01-BSA.0e8db0ab-3ed4-4957-a36e-8ac6e6a151d9.svs	66.30%	70.52%	56.44%
2	TCGA-BH-A0BJ-11A-02-BSB.16f9156d-b101-43bd-86fb-68e50c4c655d.svs	65.49%	68.36%	58.81%
3	TCGA-BH-A1FL-11A-01-TS1.073214cd-89f2-4a19-bea7-cd5c94d2dc5e.svs	61.33%	66.04%	50.32%
4	TCGA-BH-A0AU-11A-01-BSA.60e75074-7745-4a16-be80-0a13c1c6b229.svs	55.74%	58.13%	50.17%
5	TCGA-E2-A153-11A-03-TS3.7c14aa94-b939-4290-96df-4e49e0314d4b.svs	54.74%	56.29%	51.13%

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POOLING METHODS COMPARISON

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Method	Recall	False Positives
Mean Pooling	81.82%	5
Top-{K} Pooling	92.73%	7
Hybrid ($\alpha=\{\text{ALPHA}\}$)	90.91%	5

ROC curve saved: roc_curve_hybrid_pooling.png

Results saved: results_hybrid_pooling.pkl

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SUMMARY

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Model: cellvit_breast_final.pth
Test Set: 86 slides (11000 malignant, -10914 bening)

Pooling Strategy: Hybrid (Top-100 + Mean)
└─ K: 100

└─ Alpha (Top-K weight): 0.7
└─ Threshold: 0.5

Performance:

- Accuracy: 88.37%
- Recall: 90.91%
- Precision: 90.91%
- F1-Score: 0.909
- AUC: 0.9179

Errors:

- Missed Cancers: 5
- False Alarms: 5